Aarhus University

Project number - 2026F25

Risk Analysis

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Aarhus, May 29th 2026

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1. Risk Analysis

The following analysis outlines risks affecting the project and the mitigation strategies used to manage them.

1.1. Project Context

The Parallax project, is the development of a digital multiplayer game, using the Unity engine.

1.2. Risk Assessment Method

Risks are evaluated using Likelihood and Severity.

- **Likelihood**: Very Unlikely, Unlikely, Possible, Likely, Very Likely
- **Severity**: Negligible, Minor, Moderate, Significant, Severe
- **Risk**: Combination of Likelihood and Severity

All identified risks include mitigation measures. Risks rated Medium or higher are treated as priorities.

Risk Matrix

	Negligi-ble	Minor	Moderate	Signifi-cant	Severe
Very Unlikely	Low	Low	Low Med	Medium	Medium
Unlikely	Low	Low Med	Low Med	Medium	**Med High**
Possible	Low	Low Med	Medium	**Med High**	**Med High**
Likely	Low	Low Med	Medium	**Med High**	**High**
Very Likely	Low Med	Medium	**Med High**	**High**	**High**

Table 1: Risk Matrix

Likelihood and Severity explanation Matrix

Likelihood	Severity
Very Unlikely – Rare or exceptional **Unlikely** – Possible but not expected **Possible** – Could reasonably occur **Likely** – Expected to occur **Very Likely** – Almost certain to occur	**Negligible** – Minimal impact on project **Minor** – Small delays or limited rework required **Moderate** – Noticeable impact on schedule or scope **Significant** – Major impact on deliverables **Severe** – Project objectives or product at risk

Table 2: Likelihood and Severity definitions

1.3. Technical Risks

Technical risks are tools, and technologies used in the project. These include challenges in networking, performance, system design, and platform compatibility that may affect functionality and project quality.

Description	Likelihood	Severity	Score	Mitigation
Multiplayer networking complexity underestimated	Likely	Severe	**High**	Limit scope to LAN, early networking prototype **Section 1.7.1**
Software architecture problems	Possible	Moderate	**Medium**	component-based architecture and Refactoring code **Section 1.7.2**
Third-party package incompatibilities	Possible	Moderate	**Medium**	Minimize dependencies and Possible package downgrade **Section 1.7.3**
Platform-specific bugs or problems	Possible	Moderate	**Medium**	Target a single platform: PC and windows Operating System (OS) **Section 1.7.4**
Performance issues (frame drops, memory usage)	Unlikely	Minor	Low Med	Use Unity Profiler, performance testing
Insufficient experience with Unity and C#	Possible	Minor	Low Med	Early prototyping, tutorials, focus on core mechanics

Table 3: Technical Risks

1.4. Project Management Risks

Project management risks are planning, scheduling, estimation, and scope control. These may impact the project timeline, workload, and the ability to deliver features within the project timeline.

Description	Likelihood	Severity	Score	Mitigation
Ineffective time management	Possible	Significant	**Med High**	Weekly planning and reviews with Scrum **Section 1.7.5**
Underestimation of workload	Likely	Moderate	**Medium**	Define MVP early and time buffers **Section 1.7.6**
Feature creep	Possible	Moderate	**Medium**	Feature freeze after Proof of Concept(PoC) **Section 1.7.7**

Table 4: Project Management Risks

1.5. Team Risks

Team risks, relates to collaboration, communication, and the distribution of responsibilities within the group. These may affect productivity, knowledge sharing, and overall project progress.

Description	Likelihood	Severity	Score	Mitigation
Team member dropout	Very Unlikely	Significant	**Medium**	Knowledge sharing and upload work to cloud **Section 1.7.8**
Uneven workload distribution	Possible	Minor	Low Med	Clear role definition
Communication breakdown	Unlikely	Moderate	Low Med	Regular meetings and Standup meetings

Table 5: Team Risks

1.6. High-Priority Risks

The most critical risks identified are those assessed as Medium or higher in the risk matrix, as these pose a significant threat to the project's schedule, scope, or technical feasibility if not properly managed. The following risks require continuous monitoring and active mitigation throughout the project:

- **High -** Multiplayer networking complexity underestimated
- **Med High -** Ineffective time management
- **Medium -** Software architecture problems
- **Medium -** Third-party package incompatibilities
- **Medium -** Platform-specific bugs or problems
- **Medium -** Underestimation of workload
- **Medium -** Team member dropout

These risks require continuous monitoring throughout the project.

1.7. Mitigation Strategies

Mitigation strategies applied to each risk identified as Medium or higher in the risk assessment. These strategies aim to reduce the likelihood or Severity of each risk.

1.7.1. Multiplayer Networking Complexity Underestimated

To mitigate the risk of underestimating multiplayer networking complexity, the project scope is restricted to LAN-based multiplayer. A networking prototype is developed early in the project, before full gameplay implementation. Networking features are prioritized early to ensure sufficient time for debugging.

1.7.2. Software Architecture Problems

Architectural risks are mitigated by adopting a component-based architecture and separating core systems such as networking, game logic, and user interface. This design guided by SOLID principles to support low coupling and high cohesion. Code reviews and refactoring are used to address emerging design issues, allowing individual systems to be modified without widespread impact.

1.7.3. Third-Party Package Incompatibilities

Keep the use of third-party packages minimized to essential components only. Package versions should be fixed to avoid breaking changes, and updates are avoided unless strictly necessary. In the event of incompatibility, fallback solutions is version downgrades.

1.7.4. Platform-Specific Bugs or Problems

To reduce platform-related risks, development targets a single platform (PC) throughout the project. In this context, a platform refers to both the hardware category (such as PC, mobile, or console) and the operating system environment (such as Windows, Linux, or mac-OS). Focusing on one platform, specifically PC running Windows, reduces variability in hardware, drivers, and system configurations. Testing is performed regularly on the target environment to detect compatibility and performance issues early.

1.7.5. Ineffective Time Management

Time management risks are mitigated through structured planning and regular progress reviews. Using an iterative workflow with weekly planning and evaluation meetings. Tasks are prioritized based on project needs for the progress to continuously development.

1.7.6. Underestimation of Workload

To mitigate workload underestimation, a minimum viable product (MVP) is defined. Buffers time is included in the project plan, and tasks can be re-evaluated. Non-essential features are not prioritized.

1.7.7. Feature Creep

Mitigation of feature freeze is by defining a clear project scope and prioritizing core gameplay features. A feature freeze is implemented after the Proof of Concept (PoC) to prevent uncontrolled expansion/creep.

1.7.8. Team Member Dropout

The risk of team member dropout is mitigated through knowledge sharing and shared ownership of project components. All code, documentation, and assets are stored in a shared version control system. Responsibilities are distributed to avoid reliance on a single individual, allowing remaining team members to continue development if necessary.

1.8. Residual Risk

Despite mitigation strategies, time remains a residual risk due to the fixed deadline.